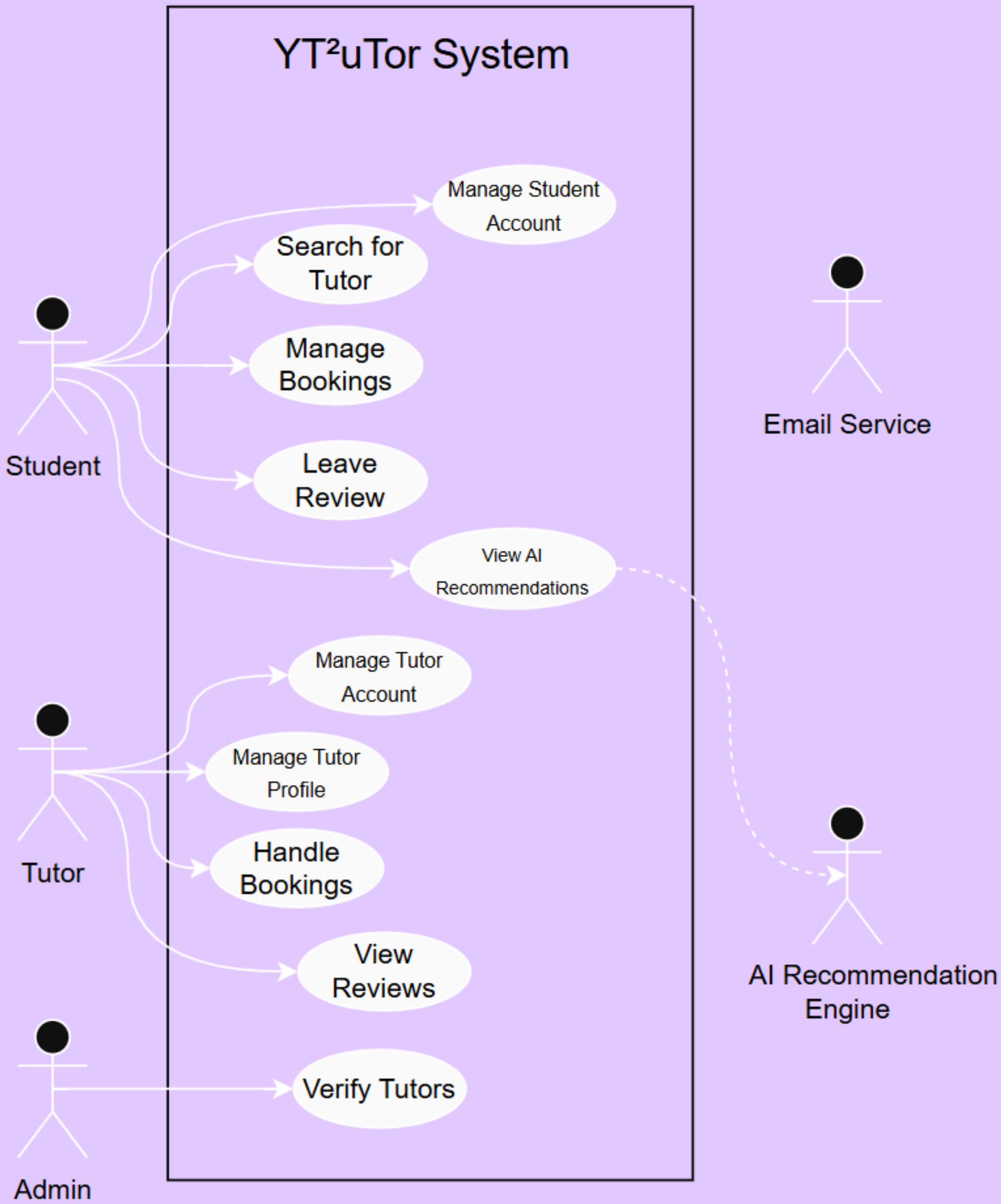
Use Case Diagram:



TEST CASES

UC-01: Manage Account (Student & Tutor)

TC ID	Title	Test Data	Steps	Expected Result
TC-01	Student registers successfully	Name: AhmadEmail: ahmad@example.comPassword: pass123User Type: Student	1. Open registration page2. Select "Student"3. Enter valid data4. Click Register	User created. Redirected to login page.
TC-02	Tutor registers successfully	Name: SaraEmail: sara@example.comPassword: pass123Bio: 5 years teaching MathUser Type: Tutor	1. Select "Tutor"2. Enter valid data with bio3. Click Register	User created with type Tutor. Bio saved.
TC-03	Register with existing email	Email: existing@example.com	1. Enter email already in database2. Click Register	Error: "Email already registered."
TC-04	Login successful	Email: ahmad@example.comPassword: pass123	1. Enter valid credentials2. Click Login	Redirected to dashboard.
TC-05	Login with wrong password	Email: ahmad@example.comPassword: wrongpass	1. Enter correct email, wrong password2. Click Login	Error: "Invalid email or password."
TC-06	Update profile	User: STU-001New name: "Ahmad Al"	1. User logs in2. Goes to profile settings3. Updates name4. Click Save	Profile updated. New name displayed.
TC-07	Change password	User: STU-001Old: pass123New: newpass456	1. User goes to change password2. Enter old and new password3. Click Save	Password changed. Can login with new password.

UC-02: Search for Tutors (Student)

TC ID	Title	Test Data	Steps	Expected Result
TC-08	Search by skill name	Skill: "Python"	1. Student enters "Python"2. Click Search	List of tutors who teach Python displayed.
TC-09	Search with no results	Skill: "UnknownSkill"	1. Enter non-existent skill2. Click Search	Message: "No tutors found."
TC-10	Search with availability filter	Day: MondayTime: 3:00 PM	1. Student selects day/time2. Click Search	Only tutors available Monday 3:00 PM shown.

UC-03: Manage Bookings (Student)

TC ID	Title	Test Data	Steps	Expected Result
TC-11	View tutor availability	Tutor: TUT-001Day: Monday	1. Student views tutor profile2. Check availability calendar	Available slots shown correctly.
TC-12	Book available slot	Tutor: TUT-001Time: Monday 3:00 PM (available)	1. Student selects available slot2. Click Book	Booking created with status "pending".
TC-13	Book unavailable slot	Tutor: TUT-001Time: Monday 5:00 PM (not available)	1. Student selects unavailable slot2. Click Book	Error: "Slot not available."
TC-14	View upcoming bookings	Student: STU-001Has 2 confirmed future bookings	1. Student goes to "My Bookings"2. View "Upcoming" tab	Upcoming sessions displayed with dates, times, tutors.
TC-15	View past bookings	Student: STU-001Has 3 completed past sessions	1. Student views "Past" tab	Past sessions displayed. "Leave Review" button visible.
TC-16	Email sent when booking confirmed	Booking: BK-001Student: STU-001	1. Tutor confirms booking2. Wait 1 minute	Student receives email: "Your booking with [Tutor] is confirmed."
TC-17	Email sent when booking rejected	Booking: BK-002Student: STU-002	1. Tutor rejects booking2. Wait 1 minute	Student receives email: "Your booking request was rejected."

UC-04: Leave Review (Student)

TC ID	Title	Test Data	Steps	Expected Result
TC-18	Leave review with comment	Booking: BK-001 (completed)Rating: 5Comment: "Great tutor!"	1. Student clicks Leave Review2. Select rating, enter comment3. Submit	Review saved. Tutor rating updated.
TC-19	Leave review without comment	Booking: BK-002 (completed)Rating: 4Comment: (empty)	1. Submit without comment	Review saved. Comment optional.
TC-20	Cannot review before session	Booking: BK-003 (pending, not completed)	1. Student tries to review before session	No review button visible.

UC-05: View AI Recommendations (Student)

TC ID	Title	Test Data	Steps	Expected Result
TC-21	Natural language request	Query: "I need help with Python functions"	1. Student types natural request2. Click AI Search	Extracted: ["Python functions"] Top 3 tutors shown.
TC-22	Request with multiple skills	Query: "Help with Python and Django"	1. Student types request2. Click AI Search	Extracted: ["Python", "Django"] Tutors with either skill shown.
TC-23	Urgent request	Query: "Need help for exam tomorrow!"	1. Student types urgent request2. Click AI Search	Urgency detected. Available now tutors prioritized.
TC-24	No matching tutors	Query: "UnknownSkill"	1. Student types request2. Click AI Search	Message: "No tutors found."

UC-06: Manage Tutor Profile (Tutor)

TC ID	Title	Test Data	Steps	Expected Result
TC-25	Add skill successfully	Tutor: TUT-001Skill: Python	1. Tutor goes to "My Skills"2. Selects Python3. Click Add	Skill added to tutor profile.
TC-26	Add duplicate skill	Tutor: TUT-001Skill: Python (already added)	1. Tries to add Python again2. Click Add	Error: "Skill already added."
TC-27	Upload valid PDF certificate	File: certificate.pdf (2MB)	1. Tutor goes to Qualifications2. Select PDF file3. Click Upload	File saved. Pending verification.
TC-28	Upload invalid file type	File: image.gif	1. Select GIF file2. Click Upload	Error: "Only PDF, JPG, PNG allowed."
TC-29	Set weekly availability	Slots: Monday 3-5 PM, Wednesday 10-12 PM	1. Tutor selects days/times2. Click Save	Slots saved. Students can see them.
TC-30	Update availability	Remove Monday 3-5, add Thursday 2-4	1. Edit slots2. Click Save	Old slots removed. New slots added.

UC-07: Handle Bookings (Tutor)

TC ID	Title	Test Data	Steps	Expected Result
TC-31	View pending requests	Tutor: TUT-001Has 2 pending requests	1. Tutor goes to "Booking Requests"	List of pending requests shown.
TC-32	No pending requests	Tutor: TUT-001No pending	1. Tutor views requests	Message: "No pending requests."
TC-33	Confirm booking	Request: RQ-001 (pending)	1. Tutor clicks Confirm2. Confirm action	Status = "confirmed". Slot booked.
TC-34	Reject booking	Request: RQ-002 (pending)	1. Tutor clicks Reject2. Confirm action	Status = "rejected". Slot freed.
TC-35	Confirm slot already taken	Request: RQ-003 (slot double-booked)	1. Tutor tries to confirm	Error: "Slot no longer available."
TC-36	View upcoming sessions	Tutor: TUT-001Has 3 confirmed future sessions	1. Tutor goes to "My Schedule"	Upcoming sessions displayed.
TC-37	View session history	Tutor: TUT-001Has 10 past sessions	1. Tutor views "History" tab	Past sessions listed with dates, status.

UC-08: View Reviews (Tutor)

TC ID	Title	Test Data	Steps	Expected Result
TC-38	View average rating	Tutor: TUT-001Has 5 reviews: avg 4.8	1. View tutor profile	Shows 4.8 (5 reviews)
TC-39	View recent reviews	Tutor: TUT-001Has 3 recent reviews	1. View tutor profile	Last 3 reviews displayed with comments.

UC-09: Verify Tutors (Admin)

TC ID	Title	Test Data	Steps	Expected Result
TC-40	Verify tutor	Tutor: TUT-001 (has uploaded certificate)	1. Admin views qualification2. Click Verify	Tutor profile shows "Verified" badge.
TC-41	Reject verification	Tutor: TUT-002 (invalid certificate)	1. Admin views qualification2. Click Reject	Tutor notified. Certificate removed.

NFR-01: Search Performance

TC ID	Title	Test Data	Steps	Expected Result
TC-P01	Search results return within 3 seconds under 30 concurrent users	30 users simultaneously searching "Python"	1. Simulate 30 concurrent users2. Each user performs skill search3. Measure response time	All search requests complete within 3 seconds.

NFR-02: AI Recommendation Performance

TC ID	Title	Test Data	Steps	Expected Result
TC-P02	AI recommendations load within 4 seconds under 20 concurrent users	20 users simultaneously sending AI queries	1. Simulate 20 concurrent users2. Each user sends natural language query3. Measure response time	All AI recommendation requests complete within 4 seconds.

NFR-03: Password Security (PBKDF2 Hashing)

TC ID	Title	Test Data	Steps	Expected Result
TC-S01	Passwords are hashed using PBKDF2	User: test@example.com Password: mySecret123	1. Register new user with password2. Inspect database	Password stored as hash (not plain text). Hash algorithm is PBKDF2.
TC-S02	Passwords cannot be reversed from hash	User: test@example.com Password: mySecret123	1. Register user2. Attempt to derive password from stored hash	Cannot retrieve original password from hash.

NFR-04: Data Security (UUID and No Sensitive Data in URLs)

TC ID	Title	Test Data	Steps	Expected Result
TC-S03	User identifiers use UUID v7 instead of sequential integers	User: STU-001	1. Register new user2. Check database user ID	User ID is UUID v7 (not sequential integer like 1, 2, 3).
TC-S04	No sensitive data in URL parameters	User: STU-001Email: student@example.com	1. Log in2. Check all URLs during navigation3. Check profile update URLs	No email, password, or phone number appears in any URL.
TC-S05	User cannot access another user's profile via URL	User A: STU-001 (logged in)User B: STU-002	1. User A logs in2. Tries to access /profile/UUID-of-B3. Tries to access /profile/UUID-of-A	Access to other profile denied (error or redirect). Access to own profile allowed.
TC-S06	User cannot access another user's bookings via URL	User A: STU-001 (logged in)User B: STU-002	1. User A logs in2. Tries to access /bookings/UUID-of-B3. Tries to access /bookings/UUID-of-A	Access to other bookings denied. Access to own bookings allowed.

NFR-05: Browser Compatibility

TC ID	Title	Test Data	Steps	Expected Result
TC-C01	System works on Chrome	Browser: Chrome (latest version)	1. Open site in Chrome2. Navigate all pages3. Perform all key functions	All pages render correctly. All functions work.
TC-C02	System works on Firefox	Browser: Firefox (latest version)	1. Open site in Firefox2. Navigate all pages3. Perform all key functions	All pages render correctly. All functions work.
TC-C03	System works on Edge	Browser: Edge (latest version)	1. Open site in Edge2. Navigate all pages3. Perform all key functions	All pages render correctly. All functions work.

NFR-06: Mobile Responsiveness

TC ID	Title	Test Data	Steps	Expected Result
TC-U01	Site works on iPhone SE (375px width)	Device: iPhone SE screen size (375px)	1. Open site on mobile simulator2. Navigate all pages3. Test all interactive elements	No horizontal scroll. All buttons tappable. Text readable without zoom.
TC-U02	Site works on iPhone 12 (390px width)	Device: iPhone 12 screen size (390px)	1. Open site on mobile simulator2. Navigate all pages3. Test all interactive elements	No horizontal scroll. All buttons tappable. Text readable without zoom.
TC-U03	Site works on Android (360px width)	Device: Android screen size (360px)	1. Open site on mobile simulator2. Navigate all pages3. Test all interactive elements	No horizontal scroll. All buttons tappable. Text readable without zoom.
TC-U04	Forms usable on mobile	Device: Any mobile (375px)	1. Open registration form on mobile2. Fill and submit form3. Open search form	Input fields visible. Keyboard appears. Form submits successfully.

NFR-07: File Upload Validation

TC ID	Title	Test Data	Steps	Expected Result
TC-U05	Upload file exceeds size limit	File: 6MB PDF	1. Tutor uploads 6MB file2. Click Upload	Error: "File too large. Max 5MB." File not saved.
TC-U06	Upload valid PDF file	File: 2MB PDF	1. Tutor uploads 2MB PDF2. Click Upload	File saved successfully.
TC-U07	Upload valid JPG image	File: 1MB JPG	1. Tutor uploads 1MB JPG2. Click Upload	File saved successfully.
TC-U08	Upload valid PNG image	File: 1MB PNG	1. Tutor uploads 1MB PNG2. Click Upload	File saved successfully.
TC-U09	Upload invalid file type (GIF)	File: image.gif	1. Tutor uploads GIF file2. Click Upload	Error: "Only PDF, JPG, PNG allowed." File not saved.
TC-U10	Upload invalid file type (DOCX)	File: document.docx	1. Tutor uploads DOCX file2. Click Upload	Error: "Only PDF, JPG, PNG allowed." File not saved.
TC-U11	Upload file with no extension	File: "certificate" (no extension)	1. Tutor uploads file without extension2. Click Upload	Error: "Invalid file type." File not saved.

NFR-08: System Scalability

TC ID	Title	Test Data	Steps	Expected Result
TC-SC01	System handles 50 concurrent users	50 concurrent users performing mixed operations	1. Simulate 50 users2. Mix of search, booking, login operations3. Monitor system stability	System remains responsive. No crashes. All operations complete.

NFR-09: Email Delivery Reliability

TC ID	Title	Test Data	Steps	Expected Result
TC-R01	Booking confirmation email sent within 5 minutes	Booking: BK-001Student: STU-001	1. Tutor confirms booking2. Start timer3. Wait for email	Email received within 5 minutes.
TC-R02	Booking rejection email sent within 5 minutes	Booking: BK-002Student: STU-002	1. Tutor rejects booking2. Start timer3. Wait for email	Email received within 5 minutes.
TC-R03	Password reset email sent within 5 minutes	User: STU-001Email: student@example.com	1. User requests password reset2. Start timer3. Wait for email	Email received within 5 minutes.

NFR-10: System Availability

TC ID	Title	Test Data	Steps	Expected Result
TC-A01	Site accessible 24/7 during evaluation	URL: YT²uTor deployed URL	1. Access site at different times (day/night)2. Test across multiple days	Site accessible. No downtime during evaluation period.